

MAE 143 A - Fall 2019

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Homework 2

1. Exercise 2.3 from Boulet FSS
2. Exercise 2.5 from Boulet FSS
3. Exercise 3.10 from Boulet FSS
4. Exercise 3.11 from Boulet FSS
5. Are the linear systems in problems 1-4 BIBO stable? Explain.
6. As in continuous-time, a discrete-time step is the function

$$1[n] = \begin{cases} 1, n \geq 0, \\ 0, n < 0 \end{cases}.$$

What is the relationship between $1[n]$ and the discrete-time impulse

$$\delta[n] = \begin{cases} 1, n = 0, \\ 0, n \neq 0 \end{cases}.$$

7. Assume that someone has calculated the step-response of a discrete-time linear time-invariant system, that is the signal

$$y_1[n] = \sum_{k=-\infty}^{\infty} g[k]1[n-k]$$

Use the solution to p6 to show how to calculate $g[n]$ from $y_1[n]$.

8. Calculate the step response to the discrete-time linear time-invariant system

$$y[n] = u[n] - u[n-1].$$

Such a system is used, often with much peril, to approximate the derivative of a sampled signal u . Use the solution to p7 to calculate the corresponding $g[n]$ from the step response. Verify that your answer is correct by directly calculating the impulse response.